23 **L** is the circle with equation $x^2 + y^2 = 4$

$P\left(\frac{3}{2}, \frac{\sqrt{7}}{2}\right)$ is a point on **L**.

Find an equation of the tangent to **L** at the point P.

The circle centre is O(0, 0).

Gradient of OP = $\frac{\sqrt{7}}{2} \div \frac{3}{2} = \frac{\sqrt{7}}{3}$

1 mark
Use rise ÷ run from the centre to P

The tangent is perpendicular to the radius OP.

Tangent gradient = $-1 \div \frac{\sqrt{7}}{3} = \frac{-3}{\sqrt{7}}$

1 mark
Use the negative reciprocal

Use $y - y_1 = m(x - x_1)$:

$$y - \frac{\sqrt{7}}{2} = \frac{-3}{\sqrt{7}}\left(x - \frac{3}{2}\right)$$

Multiply both sides by $2\sqrt{7}$:

$$2\sqrt{7}\,y - 7 = -6x + 9$$

(+6x +7) (+6x +7)

$$6x + 2\sqrt{7}\,y = 16$$

(÷2) (÷2)

$$3x + \sqrt{7}\,y = 8$$

1 mark
Correct tangent equation in any equivalent form

$3x + \sqrt{7}\,y = 8$

(Total for Question 23 is 3 marks)

TOTAL FOR PAPER IS 80 MARKS

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